

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

PAPER 5 Practical Test

9189/5

JUNE 2016 SESSION

1 hour 20 minutes

Candidates answer on the question paper.

Additional materials:

as listed in Instructions to Supervisors.

Electronic calculator

TIME 1 hour 20 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 7 printed pages and 1 blank page.

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1

(a)

You are required to determine the percentage of Fe^{2+} in a fertiliser sample by a titration method, given that acidified MnO_4^- oxidises only nitrite ions in the sample.

FA1 is the fertiliser sample.

FA3 is $1.5 \text{ g dm}^{-3} \text{ KMnO}_4$

- (i) Weigh between 1.90 g and 2.30 g of FA1, using a 100 cm^3 beaker.

Record the weighings in Table 1.1.

Table 1.1

mass of a 100 cm^3 beaker + FA1 / g	
mass of an empty 100 cm^3 beaker / g	
mass of FA1 / g	

[3]

- (ii) Dissolve the FA1 by adding distilled water into the beaker and stir with a glass rod.

Transfer the solution into a 250 cm^3 volumetric flask.

Rinse the beaker with distilled water and add the washings into the volumetric flask.

Top up to the mark with distilled water and label the solution FA2.

Pipette 25 cm^3 of FA3 into a conical flask.

Add 10 cm^3 of 1.0 mol dm^{-3} sulphuric acid.

Titrate the FA3 against FA2 until the solution is decolourised.

Repeat the titration as many times as necessary to obtain accurate results.

Record the titration results in Table 1.2.

Examiner's
Use

Table 1.2

final burette reading/cm ³					
initial burette reading/cm ³					
volume of FA2 used / cm ³					

Summary

_____ cm³ of FA3 required _____ cm³ of FA2

to completely oxidise the nitrite ions.

[15]

- (b) Explain why the colour changes from purple to colourless at end point?

[1]

- (c) Calculate the

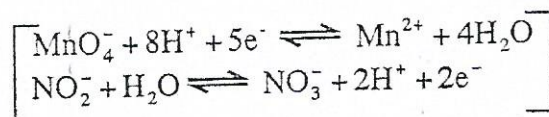
- (i) concentration of $\text{KMnO}_{4(\text{aq})}$ in mol dm^{-3} ,
[A_r: K = 39.1, Mn = 54.9 and O = 16.0]

[1]

- (ii) number of moles of $\text{MnO}_{4(\text{aq})}^-$ in 25 cm³ FA3.

[1]

- (d) (i) Write a balanced chemical equation for the redox reaction taking place.



[1]

- (ii) Calculate the number of moles of NO_2^- in the titre.

[1]

- (iii) Hence calculate the concentration of NO_2^- in FA2.

[2]

- (e) Calculate the percentage by mass of KNO_2 in the fertiliser sample.
[A_r : N = 14.]

[3]

(f) Normally $\text{KMnO}_{4(\text{aq})}$ is run from the burette into the titrant.

Suggest a reason why that could not be done in this experiment.

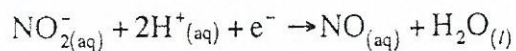
[1]

[Total: 29]

ASSESSMENT OF PLANNING SKILLS

DO NOT CARRY OUT YOUR PLAN

- 2 Nitrites react with nitric acid according to the following equation:



Design an experiment to verify the percentage composition of potassium nitrite, KNO_2 , in the fertilizer sample used in **question 1**.

Assuming you are provided with

apparatus used in **question 1**

50 cm³ of a 0.2 mol dm⁻³ $\text{HNO}_{3(\text{aq})}$

100 cm³ of a 0.1 mol dm⁻³ $\text{NaOH}_{(\text{aq})}$

phenolphthalein indicator

teat pipette

Present your plan as a sequence of numbered steps.

The plan